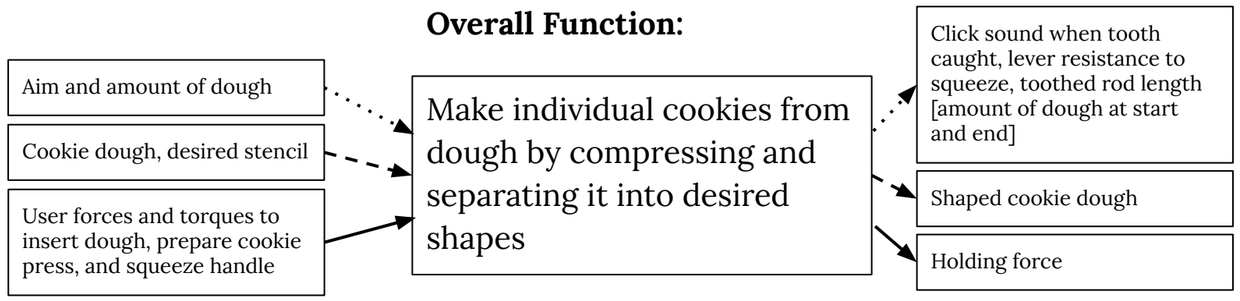
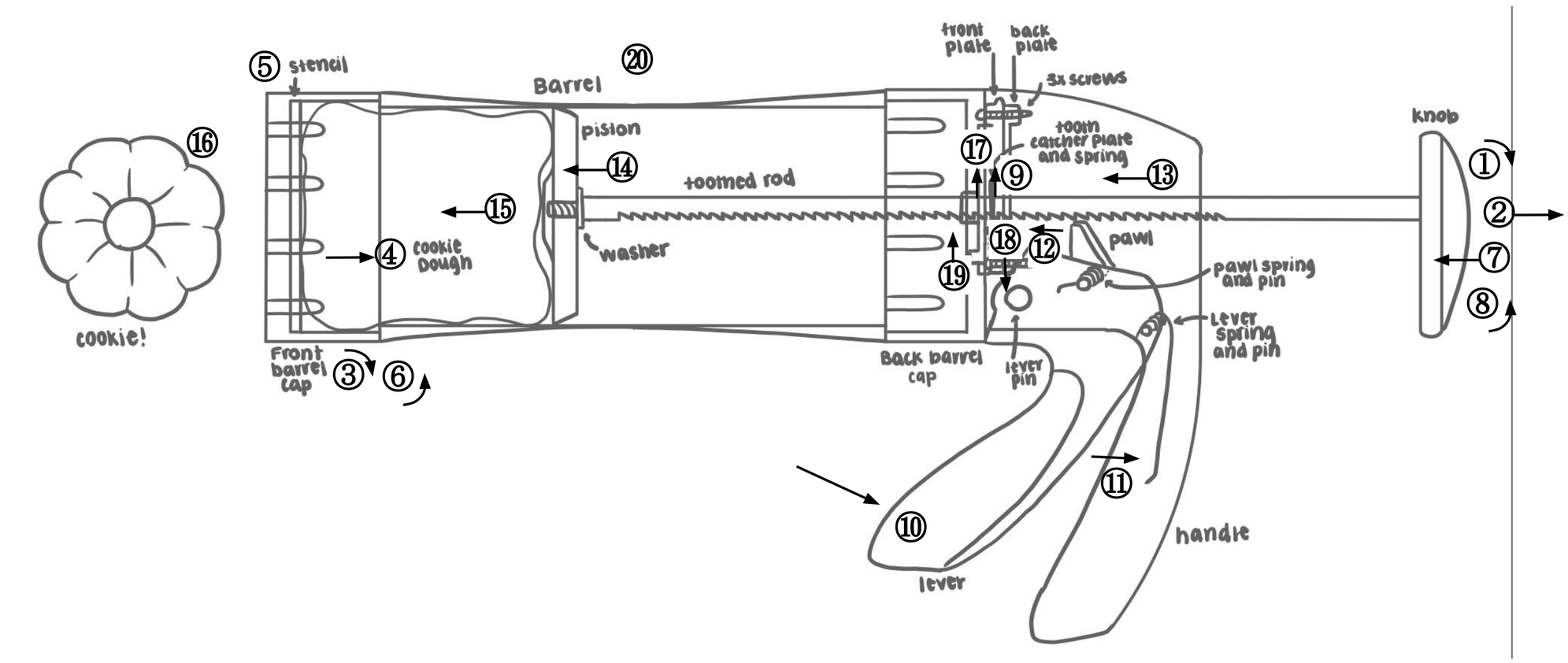
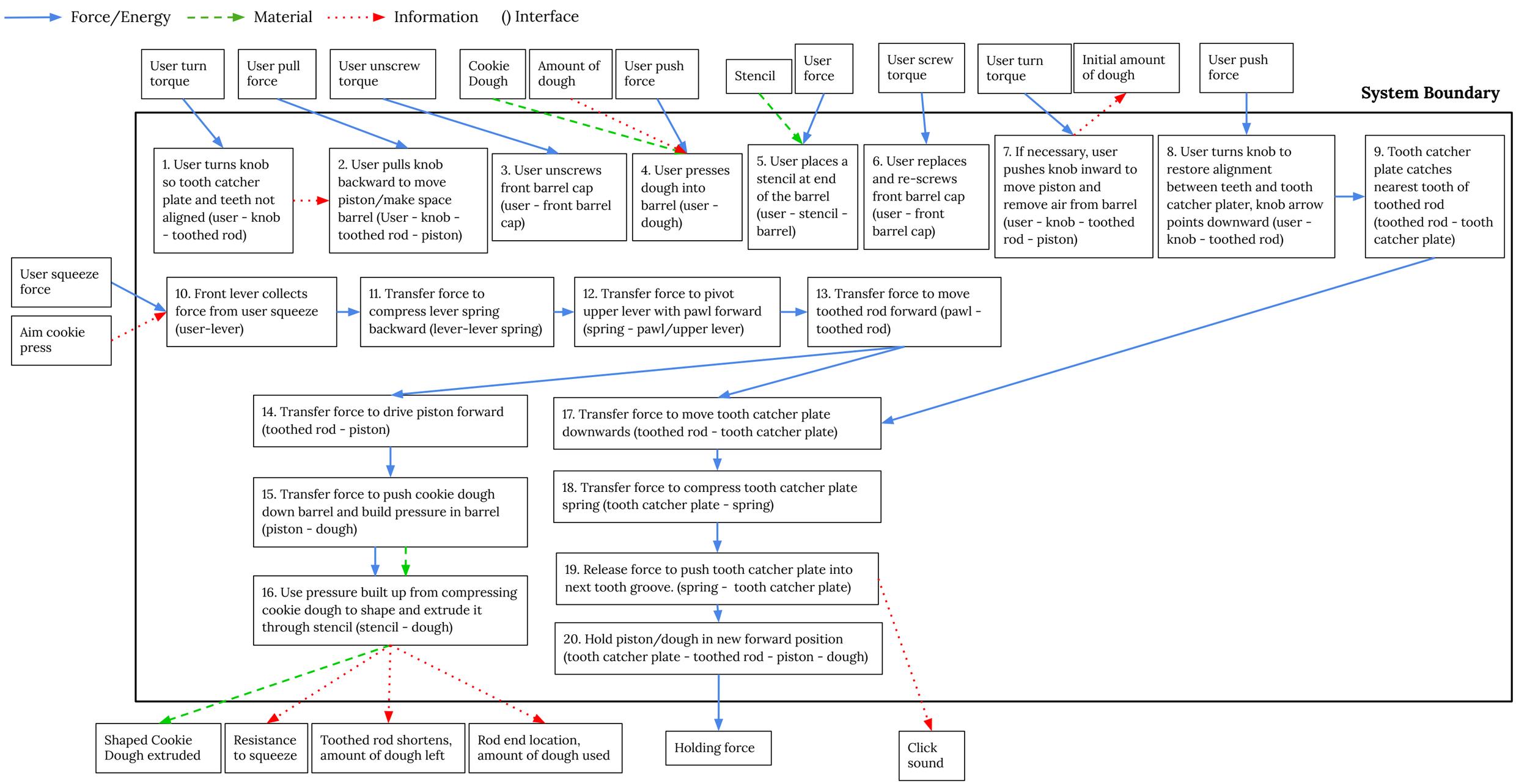
Overall Function:



Sketch:



Functional Decomposition Diagram:



Product Teardown Write-Up

Relating Pawl Engagement Angle & Ratchet Tooth Size with Linear Dough Displacement

Ani Gregori Asadourian and Anne Jumper

One design parameter that is related to the performance of the hand-squeeze cookie press is the size of the teeth on the toothed rod and the angle of the pawl. This product is a linear ratchet and pawl mechanism, with the ratchet being on the rod, and the pawl being a metal piece engaged by the hand lever. Ultimately, the size of the teeth on the ratchet and the angle of the pawl needed to push the pawl to the next tooth affect the force the user needs to apply to the lever in order to engage the mechanism. This also affects the output force of the piston onto the cookie dough, which would ultimately affect the size of the cookies by determining how much dough is extruded through the stencil with each click of the ratchet.

Since this ratchet mechanism is linear, an overall equation to describe the linear force on the system is $F_{forward} = F_{applied} - F_{friction}$. The ratchet has a coefficient of friction that acts against the applied force to inhibit forward motion, and thus if the force applied by the user is greater than the force of friction, the pawl can move to the next tooth. If the size of the teeth were decreased, the pawl would need to be moved at a smaller angle to get to the next tooth and thus over a smaller distance, which decreases the amount of friction between the pawl and ratchet and thus requires less of an applied force to move the rod forward. The recoil force on the mechanism can be described as $F_{backward} = F_{applied} * cos(\theta)$, where theta is the angle the pawl needs to move to get to the next tooth (engagement angle). A ratchet mechanism only works in one direction, where the rod cannot move backward. If the size of the tooth is decreased, theta would decrease and thus to keep the backward force on the teeth the same, the applied force decreases.

Decreasing the size of the teeth might then seem beneficial to the user, however, there are downsides to this choice. Firstly, smaller teeth on a ratchet mechanism can cause functional issues as debris could get caught in the teeth and make it difficult to click from one tooth to the next. Further, the normal force acting on the pawl can be described as $N = F_{tangential} * cos(\theta)$,

where theta is the pawl angle and the friction force can then be described as $F_f = \mu * N$. Therefore, changing the angle of the pawl affects the frictional force acting on the pawl and thus the frictional force needed to move the pawl to the next tooth. If the tooth size is decreased, the angle is decreased and thus more friction is created between the pawl and the teeth, which could cause wear and deformation over time, perhaps breaking the small teeth. (More friction is needed to push the pawl along the finer, more complex ratchet teeth). Moreover, if the teeth are smaller and thus there is less distance between them, there will be less displacement of the piston for each click of the ratchet, as the distance the ratchet moves is proportional to the size of the teeth and equal to the distance the piston moves. Thus, the piston would extrude less dough as the distance the piston moves is proportional to the dough extruded through the stencil, which would create smaller cookies. However, smaller teeth do allow for finer adjustments and less backlash/recoil, which in this case would give the user more control over the cookie size. Overall, the designers and manufacturers of this cookie press had to consider these tradeoffs to decide on the size of the teeth and pawl engagement angle that minimizes the needed user input force to ultimately output their desired cookie size while providing mechanical advantage.